Panel A

Human Perception



Question: Which image looks safer for bicycle users?

Human Votes

Data cleaning: Filter bad votes

VLM Perception



Prompt: Which image looks safer for bicycle users?

GPT ranking Votes

TrueSkill scoring

Human ranking

GPT Ranking

Human-AI alignment score

Panel B

Sample images



Computer vision feature extraction

Models

Prediction

Panoptic segmentation:
- Global
- Grid

Object detection:
- Probability
- Count
- Area

Image statistics

Composite features

Deeper MLP (128→128→64→64)

XGBoost (Tuned)

CatBoost (Tuned)

LightGBM (Tuned)

Low = Least safe

Medium = Moderate

High = Safest